

TEST PLAN

MaharaTech E-Learning Platform

<https://maharatech.gov.eg/course/index.php>

Document Title	MaharaTech Test Plan
Version	1.0
Prepared By	QA Team (5 Members)
Date	2025
Status	Draft

1. Introduction

This document describes the Test Plan for the MaharaTech E-Learning Platform, accessible at <https://maharatech.gov.eg/course/index.php>. MaharaTech is an Egyptian government-backed e-learning platform that offers a wide range of technical and professional courses to learners across Egypt and the Arab world.

The purpose of this test plan is to define the testing strategy, scope, objectives, resources, schedule, and acceptance criteria to ensure the platform functions correctly, provides a good user experience, and meets quality standards.

1.1 Purpose

This test plan aims to:

- Define acceptance criteria for all in-scope features of the platform.
- Provide structured test scenarios to guide exploratory and scripted testing.
- Identify and document defects before final delivery.
- Ensure the platform is usable on various screen sizes and browsers.
- Distribute testing work fairly among 5 team members.

1.2 Project Scope

The following modules and features are included in the test scope:

- User Registration & Login (authentication)
- Course Catalog — browsing, searching, filtering, sorting
- Course Enrollment and access
- Course Content consumption — video lectures, materials
- Quiz & Assessment functionality
- Certificate generation and download
- User Profile management
- UI/UX — responsiveness, cross-browser, accessibility, RTL Arabic support

1.3 Out of Scope

- Backend / server-side infrastructure testing
- Payment gateway testing (if applicable)
- Instructor / admin panel functionality
- Security penetration testing
- Load / stress testing

2. Test Objectives

1. Validate that all in-scope features satisfy their acceptance criteria.
2. Ensure consistent user experience across devices (mobile, tablet, desktop).
3. Verify that the platform supports Arabic RTL layout correctly.
4. Confirm that enrolled users can access course content without interruption.
5. Validate that quizzes, scoring, and certificate generation work accurately.
6. Identify and log all defects for resolution before project closure.

3. Test Team & Responsibilities

The QA team consists of 5 members. Each member is responsible for a specific set of modules to ensure full coverage and accountability.

#	Member Name	Role	Responsibilities	Modules Assigned
1	Member 1	Test Lead	Test plan creation, coordination, reporting	Registration, Login, Authentication
2	Member 2	Tester	Functional & UI testing of catalog	Course Catalog, Search & Filters
3	Member 3	Tester	Enrollment, content & learning flow testing	Enrollment, Course Content, Learning
4	Member 4	Tester	Quiz, assessment, and certification testing	Quiz / Assessments, Certificates
5	Member 5	Tester / UX QA	Profile, UI/UX, responsiveness, accessibility	User Profile, UI/UX, Cross-browser

4. Test Environment

4.1 Browsers

- Google Chrome (latest stable version)
- Mozilla Firefox (latest stable version)
- Microsoft Edge (latest stable version)

4.2 Devices / Viewports

- Desktop: 1920x1080, 1366x768
- Tablet: 768x1024 (iPad)
- Mobile: 375x667 (iPhone SE), 360x800 (Android)

4.3 Operating Systems

- Windows 10 / 11
- macOS (if available)
- Android / iOS (for mobile testing)

4.4 Test Data

- Valid and invalid email addresses
- Test user accounts (one per tester)
- Sample course enrollments pre-configured for testing

5. Test Approach

5.1 Testing Types

- Functional Testing: Verify each feature works as per acceptance criteria.
- UI/UX Testing: Validate visual appearance, layout, and usability.
- Regression Testing: Re-test fixed defects to confirm resolution.
- Compatibility Testing: Ensure consistent behavior across browsers and devices.
- Exploratory Testing: Use the provided scenarios as a thinking framework to uncover unscripted issues.

5.2 Entry & Exit Criteria

Entry Criteria

- Test environment is set up and accessible.
- Test accounts are created and verified.
- All team members have reviewed this test plan.

Exit Criteria

- All acceptance criteria have been verified for each module.
- All High priority defects are resolved or have approved workarounds.
- Test summary report is completed and signed off.

6. Test Schedule

Phase	Activity	Duration	Responsible
1	Test Plan Review & Sign-off	Day 1	All Members
2	Test Environment Setup	Day 1–2	All Members
3	Test Execution (Scenarios & AC Verification)	Day 3–7	Members 1–5
4	Defect Logging & Reporting	Day 3–7 (ongoing)	All Members
5	Defect Fix Verification / Regression	Day 8–10	Members 1–5
6	Final Test Summary Report	Day 11–12	Member 1 (Test Lead)

7. Acceptance Criteria

The following acceptance criteria define the conditions that must be met for each module to be considered successfully tested and ready for sign-off. These criteria are used by testers as the primary benchmark to determine Pass or Fail during test execution.

#	Module	Acceptance Criteria	Verification Method
AC-01	Registration	A new user can successfully register by providing a unique email address, a valid password (min. 8 chars), and required profile fields. A confirmation message is shown upon success.	Verify registration form submits successfully with valid data. Check that duplicate email triggers a clear error message. Confirm validation messages appear for empty required fields.
AC-02	Login / Auth	A registered user can log in using valid credentials and is redirected to their dashboard. Invalid credentials must produce a descriptive error without exposing security details.	Attempt login with correct and incorrect credentials. Confirm redirect on success. Verify error message on failure does not reveal whether email or password was wrong.
AC-03	Password Reset	A user who has forgotten their password can request a reset link via email. The link must expire after a defined time and allow the user to set a new valid password.	Trigger password reset flow. Confirm email is received. Verify link expires and cannot be reused. Confirm new password allows login.
AC-04	Course Catalog	All published courses must be visible on the catalog page with title, category, difficulty level, and thumbnail. Search and filter operations must return accurate, relevant results.	Browse the full catalog. Run keyword searches. Apply category and difficulty filters. Confirm results match applied criteria. Verify pagination works correctly.
AC-05	Course Enrollment	An authenticated user can enroll in any free course in one click. The enrolled course must immediately appear in the user's dashboard under 'My Courses'.	Enroll in a course as a logged-in user. Confirm instant dashboard appearance. Attempt enrollment as a guest — verify redirect to login.
AC-06	Course Content	Enrolled users can access all course materials (videos, PDFs, resources). Video playback must be smooth, and progress must be accurately tracked and persisted between sessions.	Open each content type within a course. Verify video plays without errors. Watch a portion, leave the page, and return to confirm progress is preserved.
AC-07	Quiz & Assessment	Quizzes must load all questions, enforce any time limits, and calculate scores accurately. Results and correct answers must be displayed immediately after submission.	Complete a quiz fully and partially. Verify auto-submission on timer expiry. Confirm score matches correct answers. Verify retake behavior if applicable.
AC-08	Certificate	A certificate must be automatically generated upon 100% course completion and passing any required quiz. The certificate must be downloadable as a valid PDF with the learner's name and course title.	Complete all modules of a course. Verify certificate appears in profile. Download the PDF and confirm it contains correct learner name, course title, and date.

#	Module	Acceptance Criteria	Verification Method
AC-09	User Profile	Users can view and edit their profile information (name, bio, avatar). Password changes take effect immediately and require re-login. Profile shows accurate progress across all enrolled courses.	Edit profile fields and save. Confirm changes are reflected. Change password and verify old one no longer works. Check progress section for accuracy.
AC-10	UI / Responsiveness	All pages must render correctly on mobile (360px), tablet (768px), and desktop (1366px+) without horizontal scrolling or broken layouts. Arabic RTL mode must mirror the entire UI correctly.	Use browser DevTools to simulate each viewport. Check all critical pages. Switch to Arabic and verify full RTL layout rendering on catalog, course, and profile pages.

8. Test Scenarios

The following scenarios are provided as a structured thinking framework for the QA team. They describe real-world user journeys and include exploratory questions to guide testers in discovering edge cases, usability issues, and unexpected behaviors. Testers are encouraged to think beyond the scenario description and investigate the questions raised.

Note: These scenarios are not step-by-step test cases. They are starting points for testing thinking. When executing, testers should document what they find, log any defects, and record their observations in the shared defect log.

#	Module	Scenario Title	Scenario Description	Think About / Questions to Explore
S-01	Registration	New User Completes Registration	A first-time visitor opens the site, fills in the registration form, and creates an account.	• What happens if the email format is invalid? • What if the password is too short? • Is there a CAPTCHA or bot protection? • Is the confirmation immediate or via email?
S-02	Registration	Duplicate Account Attempt	A user tries to register with an email address that is already linked to an existing account.	• Does the system reveal that the email exists (security concern)? • Is the error message helpful but not exposing? • Can the user be guided to login or password reset?
S-03	Login	Returning User Logs In	An existing user enters their credentials and expects to be taken to their personal dashboard.	• Is the redirect to the correct page (dashboard vs. course list)? • Are there any session timeout policies? • What is the lockout policy after multiple failed attempts?
S-04	Course Catalog	Learner Searches for a Specific Course	A learner uses the search bar to find a course by keyword and then filters by category.	• Does the search handle partial words or typos? • Are results ranked by relevance? • What happens if no results are found — is there a helpful empty state?
S-05	Course Catalog	Browsing Courses Without an Account	An anonymous visitor browses the course catalog and views a course detail page.	• Can the full course description be seen without login? • Is there a clear call-to-action to register or login? • Is there any content that requires login to view?

#	Module	Scenario Title	Scenario Description	Think About / Questions to Explore
S-06	Enrollment	User Enrolls and Immediately Starts Learning	An authenticated learner clicks Enroll on a course and then immediately opens the first lesson.	• Is there any delay between enrollment and access? • Does the course show as In Progress on the dashboard? • Is the first lesson automatically opened or does the user navigate manually?
S-07	Course Content	Learner Resumes a Course After Break	A learner who previously watched 3 of 10 lessons returns to the course the next day.	• Is the last-watched lesson highlighted? • Does the video resume from where they stopped, or from the beginning? • Is the progress percentage correct?
S-08	Quiz	Learner Takes a Timed Quiz	A learner starts a quiz that has a time limit and is still answering when the timer runs out.	• Is the remaining time visible at all times? • Is there a warning before time runs out (e.g., 1 minute left)? • Are partial answers submitted automatically or is everything lost?
S-09	Quiz	Learner Fails and Wants to Retry	A learner scores below the passing threshold and sees their results page.	• Is the passing score clearly communicated before the quiz? • Are correct answers shown after failure? • How many retakes are allowed? Is there a waiting period?
S-10	Certificate	Learner Earns and Downloads Certificate	After completing all course content and passing the final quiz, the learner receives a certificate.	• Is the certificate trigger automatic or does the learner request it? • Does the PDF contain the correct name, course, and date? • Can the certificate be shared or verified externally?
S-11	Profile	Learner Reviews Their Own Progress	A learner visits their profile to review which courses they have enrolled in and how far along they are.	• Is progress shown as a percentage or lesson count? • Are completed courses visually distinct from in-progress ones? • Can the learner re-access a completed course?

#	Module	Scenario Title	Scenario Description	Think About / Questions to Explore
S-12	UI / RTL	Arabic-Language User Navigates the Platform	A user switches the interface language to Arabic and navigates the course catalog and a course detail page.	• Does the entire layout flip to right-to-left? • Are buttons, icons, and navigation correctly mirrored? • Is all text properly translated, or are there English fallbacks? • Do dropdowns and modals also respect RTL?
S-13	Cross-Browser	Tester Validates Consistent Behavior Across Browsers	The same core user flow (login → enroll → watch lesson) is performed in Chrome, Firefox, and Edge.	• Are there any layout differences between browsers? • Does video playback work equally in all browsers? • Are fonts and spacing consistent? • Do any JS-heavy components behave differently?
S-14	Mobile	Learner Accesses Platform on a Smartphone	A user opens the site on a 360px Android device and tries to find and enroll in a course.	• Is the navigation menu accessible (hamburger or drawer)? • Are buttons large enough to tap comfortably? • Does the video player work on mobile? • Is text readable without zooming?

9. Defect Tracking & Reporting

9.1 Defect Severity Levels

- Critical: System crash or complete feature failure blocking testing.
- High: Major feature broken, no workaround available.
- Medium: Feature partially broken, workaround exists.
- Low: Minor UI issues, cosmetic defects.

9.2 Defect Status Definitions

- New: Defect just reported.
- Open: Defect confirmed and assigned for fixing.
- Fixed: Developer has resolved the defect.
- Retest: Awaiting tester verification.
- Closed: Defect verified as fixed.
- Rejected: Not a defect; expected behavior.

9.3 Defect Log Template

Testers should log defects using the format below. A shared defect log spreadsheet should be maintained by the team:

Bug ID	Severity	Priority	Module	Status	Description
BUG-001					(To be filled during test execution)

10. Risks & Mitigation

Risk	Likelihood	Mitigation
Platform instability / downtime	Medium	Schedule testing in off-peak hours; maintain backup access
Test accounts locked or expired	Low	Create multiple accounts; coordinate with platform admin
Team member unavailability	Medium	Cross-train members; redistribute tasks if needed
Incomplete requirements/documentation	Medium	Use live platform as reference; document assumptions
Browser / device compatibility issues	High	Test on all 3 major browsers; use BrowserStack if needed

11. Test Deliverables

- This Test Plan document (v1.0)
- Defect Log spreadsheet (maintained throughout execution)
- Test Summary Report (after execution is complete)
- Screenshots / screen recordings of critical defects

12. Approval & Sign-off

Name	Role	Signature	Date
Member 1	Test Lead		
Member 2	Tester		
Member 3	Tester		
Member 4	Tester		
Member 5	Tester / UX QA		

— End of Test Plan Document —